# $p_T$ resolution fit vs $|\eta^{truth}|$: $r_0$ free vs $r_0 = 0$

$\sigma_{p_T}/p_T = \sqrt{r_0^2/p_T^2 + r_1^2 + (r_2 \times p_T)^2}$ -- $\sigma_{68} = (q_{84} - q_{16})/2$ -- Z + Z' combined

| $\lvert\eta\rvert$ bin | N points | $r_0$ free [GeV] | $\chi^2$/ndf ($r_0$ free) | $\chi^2$/ndf ($r_0 = 0$) | $r_2$ [$GeV^{-1}$] |
|---|---|---|---|---|---|
| $0.0 \leq \lvert\eta\rvert < 0.1$ | 16 | $0.000 \pm 45.213$ | 0.16 | 0.14 | **$0.311 \pm 0.015$ $(\times 10^{-3})$** |
| $0.1 \leq \lvert\eta\rvert < 1.05$ | 16 | $0.000 \pm 30.900$ | 0.67 | 0.59 | **$0.192 \pm 0.011$ $(\times 10^{-3})$** |
| $1.05 \leq \lvert\eta\rvert < 1.3$ | 15 | $0.001 \pm 38.310$ | 0.29 | 0.25 | **$0.270 \pm 0.014$ $(\times 10^{-3})$** |
| $1.3 \leq \lvert\eta\rvert < 1.7$ | 14 | $0.000 \pm 45.304$ | 0.20 | 0.17 | **$0.283 \pm 0.016$ $(\times 10^{-3})$** |
| $1.7 \leq \lvert\eta\rvert < 2.5$ | 13 | $0.000 \pm 49.423$ | 0.05 | 0.04 | **$0.278 \pm 0.015$ $(\times 10^{-3})$** |
| $2.5 \leq \lvert\eta\rvert < 2.8$ | 11 | $0.404 \pm 0.347$ | 0.06 | 0.10 | **$0.430 \pm 0.024$ $(\times 10^{-3})$** |

In red: $\chi^2$/ndf > 3, unreliable fit

In blue: $r_2$, the nominal ATLAS resolution at high $p_T$